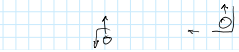
Q1.1 Kinematics (1 pt.)

Model the kinematics of the system. At the end you should have an expression of the x, y coordinates and velocities of the print head based on the motor angles θ_l and θ_r. We expect formulations for the variables:

$$x(\theta_l, \theta_r),\ y(\theta_l, \theta_r) \tag{1}$$
$$\dot{x}(\dot{\theta}_l, \dot{\theta}_r),\ \dot{y}(\dot{\theta}_l, \dot{\theta}_r). \tag{2}$$



Rotation of motor into translation

$s_l = r_w \theta_l \qquad s_r = r_w \theta_r$

If the rope moves it will result in a linear combination of x and y movements

left side will result in x and y positive

$s_L = x + y$

right side will result in x positive and y negative

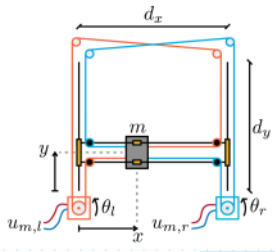
$s_R = x - y$

Add s_L and s_R for the expression for x and subtract s_R from s_L for y

$s_L + s_R = 2x \Rightarrow x = \frac{1}{2}(s_L + s_R) \qquad s_L - s_R = 2y \Rightarrow y = \frac{1}{2}(s_L - s_R)$

$\Rightarrow x(\theta_l, \theta_r) = \frac{r_w}{2}(\theta_l + \theta_r) \qquad y(\theta_l, \theta_r) = \frac{r_w}{2}(\theta_l - \theta_r)$

$\dot{x}(\dot{\theta}_l, \dot{\theta}_r) = \frac{r_w}{2}(\dot{\theta}_l + \dot{\theta}_r) \qquad \dot{y}(\dot{\theta}_l, \dot{\theta}_r) = \frac{r_w}{2}(\dot{\theta}_l - \dot{\theta}_r)$



Name	Symbol
Motor angle left	θ_l
Motor voltage left	$u_{m,l}$
Motor current left	$i_{m,l}$
Motor radius	r_w
Motor induction	L_m
Motor resistor	R_w
Motor inertia	J_w
Motor constant	K_w
Tool head position	x, y
Tool head mass	m
Nozzle voltage	u_n
Nozzle current	i_n
Nozzle capacitor	C_n
Nozzle thermal capacity	$C_{t,n}$
Nozzle thermal resistance	$R_{t,n}$
Nozzle resistor	R_n
Nozzle temperature	T_n
Environmental temperature	T_t
Bearing friction rolling	μ_r
Bearing friction linear x-direction	$\mu_{l,x}$
Bearing friction linear y-direction	$\mu_{l,y}$
Print bed temperature	T_b
Print area dimension x	d_x
Print area dimension y	d_y

$\mu_{l,x}$ accounts for both rails

$\Rightarrow \frac{1}{2}\mu_{l,x}\dot{x}^2 = \frac{1}{2}\frac{\mu_{l,x} r_w^2}{4}(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2)$

$\mu_{l,y}$ also accounts for both rails

$\Rightarrow \frac{1}{2}\mu_{l,y}\dot{y}^2 = \frac{1}{2}\frac{\mu_{l,y} r_w^2}{4}(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2)$

μ_r for each additional bearing with r_w

for two fixed orange bearings $2\left[\frac{1}{2}\mu_r\left(\frac{r_w}{r_o}\dot{\theta}_l\right)^2\right] = 2\left[\frac{1}{2}\frac{\mu_r r_w^2}{r_o^2}\dot{\theta}_l^2\right] = \frac{\mu_r r_w^2}{r_o^2}\dot{\theta}_l^2$

for two moving orange bearings $2\left[\frac{1}{2}\mu_r\left(\frac{\dot{x}}{r_o}\right)^2\right] = 2\left[\frac{1}{2}\frac{\mu_r r_w^2}{4 r_o^2}(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2)\right] = \frac{\mu_r r_w^2}{4 r_o^2}\dot{\theta}_l^2 - \frac{\mu_r r_w^2}{4 r_o^2}\dot{\theta}_l\dot{\theta}_r + \frac{\mu_r r_w^2}{4 r_o^2}\dot{\theta}_r^2$

for two fixed blue bearings $2\left[\frac{1}{2}\mu_r\left(\frac{r_w}{r_b}\dot{\theta}_l\right)^2\right] = 2\left[\frac{1}{2}\frac{\mu_r r_w^2}{r_b^2}\dot{\theta}_l^2\right] = \frac{\mu_r r_w^2}{r_b^2}\dot{\theta}_l^2$

for two moving blue bearings $2\left[\frac{1}{2}\mu_r\left(\frac{\dot{x}}{r_b}\right)^2\right] = 2\left[\frac{1}{2}\frac{\mu_r r_w^2}{4 r_b^2}(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2)\right] = \frac{\mu_r r_w^2}{4 r_b^2}\dot{\theta}_l^2 - \frac{\mu_r r_w^2}{4 r_b^2}\dot{\theta}_l\dot{\theta}_r + \frac{\mu_r r_w^2}{4 r_b^2}\dot{\theta}_r^2$

$\Rightarrow \frac{\mu_r r_w^2}{r_b^2}\left[\frac{3}{2}\dot{\theta}_l^2 - \dot{\theta}_l\dot{\theta}_r + \frac{3}{2}\dot{\theta}_r^2\right]$

→ radius for wheel bearings r_w equal for every bearing

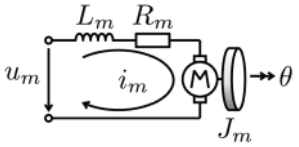
Q1.2 Dynamics (2 pts.)

Model the dynamics of the system. At the end of this part you should have a state-space description of the system with the voltages of the motors as inputs of the form

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) \tag{3}$$

with $\boldsymbol{x} = [\theta_l, \theta_r, \cdots]^\top$ and $\boldsymbol{u} = [u_l, u_r]^\top$.

AN IMPORTANT CAVEAT! Your colleague ordered the wrong motors, so instead of the typical stepper motors, you only have voltage-controlled DC motors at your disposal. As a result, you have to consider the following circuit to model the motor:



Mechanical kinetic energy

$K_{mech} = \frac{1}{2}m\left\|\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix}\right\|^2 + \frac{1}{2}J_w\dot{\theta}_l^2 + \frac{1}{2}J_w\dot{\theta}_r^2 + \frac{1}{2}m\frac{r_w^2}{4}\left(\dot{\theta}_l^2 + 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2 + \dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2\right) + \frac{1}{2}J_m\left(\dot{\theta}_l^2 + \dot{\theta}_r^2\right) = \frac{1}{2}\left(\dot{\theta}_l^2 + \dot{\theta}_r^2\right)\left(\frac{mr_w^2}{2} + J_m\right)$

Electrical kinetic energy — terms for electromagnetic coupling, don't forget sitting

$K_{elec} = \frac{1}{2}L_m\dot{q}_{m,l}^2 + \frac{1}{2}L_m\dot{q}_{m,r}^2 + K_m\dot{q}_{m,l}\dot{\theta}_l + K_m\dot{q}_{m,r}\dot{\theta}_r = \frac{1}{2}L_m\left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2\right) + K_m\left(\dot{q}_{m,l}\dot{\theta}_l + \dot{q}_{m,r}\dot{\theta}_r\right)$

→ with q being charge and therefore $\dot{q}$ equals the current

Mech. dissipative energy

$D_{mech} = \frac{1}{2}\mu_{l,x}\dot{x}^2 + \frac{1}{2}\mu_{l,y}\dot{y}^2 + \frac{\mu_r r_w^2}{r_o^2}\left[\frac{1}{2}\dot{\theta}_l^2 + \dot{\theta}_l\dot{\theta}_r + \frac{1}{2}\dot{\theta}_r^2\right] + \frac{1}{2}\frac{\mu_{l,x} r_w^2}{4}(\dot{\theta}_l^2 + 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2) + \frac{1}{2}\frac{\mu_{l,y} r_w^2}{4}(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2) + \frac{\mu_r r_w^2}{r_b^2}\left[\frac{3}{2}\dot{\theta}_l^2 + \dot{\theta}_l\dot{\theta}_r + \frac{3}{2}\dot{\theta}_r^2\right] =$

$= r_w^2\left[\underbrace{\left(\frac{\mu_{l,x}}{8} + \frac{\mu_{l,y}}{8} + \frac{3\mu_r}{2r_b^2}\right)}_{\frac{1}{r_w^2}\mu_t}\dot{\theta}_l^2 + \underbrace{\left(\frac{\mu_{l,x}}{4} - \frac{\mu_{l,y}}{4} + \frac{\mu_r}{r_b^2}\right)}_{\frac{1}{r_w^2}\mu_a}\dot{\theta}_l\dot{\theta}_r + \left(\frac{\mu_{l,x}}{8} + \frac{\mu_{l,y}}{8} + \frac{3\mu_r}{2r_b^2}\right)\dot{\theta}_r^2\right]$

Elect. dissipative energy

$D_{elec} = \frac{1}{2}R_m\dot{q}_{m,l}^2 + \frac{1}{2}R_m\dot{q}_{m,r}^2 = \frac{1}{2}R_m\left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2\right)$

External forces

for $q_{m,l}$: $u_{m,l}$

for $q_{m,r}$: $u_{m,r}$

$2\mu_t = \mu_a = r_w^2\left(\frac{\mu_{l,x}}{2} - 4\frac{\mu_r}{r_b^2}\right)$

$2\mu_t = \mu_a = r_w^2\left(\frac{\mu_{l,y}}{2} + 2\frac{\mu_r}{r_b^2}\right)$

Lagrange equation

$L = K - P = K_{mech} + K_{elec} = \frac{1}{2}\left(\dot{\theta}_l^2 + \dot{\theta}_r^2\right)\left(\frac{mr_w^2}{2} + J_m\right) + \frac{1}{2}L_m\left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2\right) + K_m\left(\dot{q}_{m,l}\dot{\theta}_l + \dot{q}_{m,r}\dot{\theta}_r\right)$

Total dissipative energy

$D = D_{mech} + D_{elec} = \mu_t\dot{\theta}_l^2 + \mu_a\dot{\theta}_l\dot{\theta}_r + \mu_t\dot{\theta}_r^2 + \frac{1}{2}R_m\left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2\right)$

to get $\frac{d}{dt}\frac{\partial L}{\partial \dot{v}_i} - \frac{\partial L}{\partial v_i} + \frac{\partial D}{\partial \dot{v}_i} = Q_i$, do partial derivatives with $v = [\theta_l, \theta_r, q_{m,l}, q_{m,r}]$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_l} = \frac{d}{dt}\left(\left(\frac{mr_w^2}{2} + J_m\right)\dot{\theta}_l\right) + \left(\frac{mr_w^2}{2} + J_m\right)\ddot{\theta}_l$	$\frac{\partial L}{\partial \theta_l} = K_m\dot{q}_{m,l}$	$\frac{\partial D}{\partial \dot{\theta}_l} = 2\mu_t\dot{\theta}_l + \mu_a\dot{\theta}_r$
$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_r} = \left(\frac{mr_w^2}{2} + J_m\right)\ddot{\theta}_r$	$\frac{\partial L}{\partial \theta_r} = K_m\dot{q}_{m,r}$	$\frac{\partial D}{\partial \dot{\theta}_r} = \mu_a\dot{\theta}_l + 2\mu_t\dot{\theta}_r$
$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_{m,l}} = \frac{d}{dt}\left(L_m\dot{q}_{m,l} + K_m\theta_l\right) = L_m\ddot{q}_{m,l} + K_m\dot{\theta}_l$	$\frac{\partial L}{\partial q_{m,l}} = 0$	$\frac{\partial D}{\partial \dot{q}_{m,l}} = R_m\dot{q}_{m,l}$
$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_{m,r}} = L_m\ddot{q}_{m,r} + K_m\dot{\theta}_r$	$\frac{\partial L}{\partial q_{m,r}} = 0$	$\frac{\partial D}{\partial \dot{q}_{m,r}} = R_m\dot{q}_{m,r}$

$\theta_l: \quad \left(\frac{mr_w^2}{2} + J_m\right)\ddot{\theta}_l - K_m\dot{q}_{m,l} + 2\mu_t\dot{\theta}_l + \mu_a\dot{\theta}_r = 0$

$\theta_r: \quad \left(\frac{mr_w^2}{2} + J_m\right)\ddot{\theta}_r - K_m\dot{q}_{m,r} + \mu_a\dot{\theta}_l + 2\mu_t\dot{\theta}_r = 0$

$q_{m,l}: \quad L_m\dot{q}_{m,l} + K_m\dot{\theta}_l + R_m q_{m,l} = u_{m,l}$

$q_{m,r}: \quad L_m\dot{q}_{m,r} + K_m\dot{\theta}_r + R_m\dot{q}_{m,r} = u_{m,r}$

$\vec{x} = \left[\theta_l, \theta_r, \dot{\theta}_l, \dot{\theta}_r, \underbrace{i_{m,l}}_{q_{m,l}}, \underbrace{i_{m,r}}_{q_{m,r}}\right]^\top \qquad \vec{u} = [u_{m,l}, u_{m,r}]^\top$

$$\dot{\vec{x}} = \begin{bmatrix} \dot{\theta}_l \\ \dot{\theta}_r \\ \frac{1}{\frac{mr_w^2}{2} + J_m}\left(K_m i_{m,l} - 2\mu_t\dot{\theta}_l - \mu_a\dot{\theta}_r\right) \\ \frac{1}{\frac{mr_w^2}{2} + J_m}\left(K_m i_{m,r} - \mu_a\dot{\theta}_l - 2\mu_t\dot{\theta}_r\right) \\ \frac{1}{L_m}\left(-K_m\dot{\theta}_l - R_m i_{m,l} + u_{m,l}\right) \\ \frac{1}{L_m}\left(-K_m\dot{\theta}_r - R_m i_{m,r} + u_{m,r}\right) \end{bmatrix}$$

$J = \frac{m r^2}{2} + J_m$

$$\dot{\vec{x}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{2\mu_1}{J} & -\frac{2\mu_2}{J} & \frac{K_m}{J} & 0 \\ 0 & 0 & -\frac{2\mu_2}{J} & -\frac{2\mu_1}{J} & 0 & \frac{K_m}{J} \\ 0 & 0 & -\frac{K_m}{L_m} & 0 & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & 0 & -\frac{K_m}{L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix} \vec{x} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix} \vec{u}$$

Q1.3 *Decouple the x and y dynamics (1 pt.)*

Investigate how you can transform the state-space model to decouple the x and y motions. You should arrive at two decoupled state-space descriptions:

$$\dot{\boldsymbol{x}}_x = f_x(\boldsymbol{x}_x, \boldsymbol{u}_x), \qquad \dot{\boldsymbol{x}}_y = f_y(\boldsymbol{x}_y, \boldsymbol{u}_y), \tag{4}$$

with $\boldsymbol{x}_x = [x, \cdots]^\top$, $\boldsymbol{x}_y = [y, \cdots]^\top$, and $\boldsymbol{u}_x$, $\boldsymbol{u}_y$ being functions of u_l and u_r.

Hint: *Review your solution from Q1.1.*

$x = \frac{r_m}{2}(\theta_l + \theta_r) \qquad y = \frac{r_m}{2}(\theta_l - \theta_r)$

$\ddot{x} = \frac{r_m}{2}\frac{1}{J}\left(I\rho\ddot{\theta}_l - \mu_1\dot{\theta}_l - \mu_2\dot{\theta}_r + K_m i_{m,l} - \mu_1\dot{\theta}_l - 2\mu_2\dot{\theta}_r + K_m i_{m,r}\right) - \frac{r_m}{2J}\left(-(\mu_1+\mu_2)\underbrace{(\dot{\theta}_l - \dot{\theta}_r)}_{=\frac{2}{r_m}\dot{x}} + K_m\underbrace{(i_{m,l} - i_{m,r})}_{=\frac{2}{r_m}i_x}\right) = -\frac{2\mu_1+\mu_2}{J}\dot{x} + \frac{K_m}{J}i_x$

$\ddot{y} = \frac{r_m}{2}\frac{1}{J}\left(I\rho\ddot{\theta}_l - \mu_1\dot{\theta}_l + K_m i_{m,l} + \mu_2\dot{\theta}_l + 2\mu_2\dot{\theta}_r - K_m i_{m,r}\right) - \frac{r_m}{2J}\left(-(\mu_1+\mu_2)\underbrace{(\dot{\theta}_l - \dot{\theta}_r)}_{=\frac{2}{r_m}\dot{y}} + K_m\underbrace{(i_{m,l} - i_{m,r})}_{=\frac{2}{r_m}i_y}\right) = -\frac{2\mu_1+\mu_2}{J}\dot{y} + \frac{K_m}{J}i_y$

$\frac{di_x}{dt} = \frac{r_m}{2}\frac{1}{L_m}\left(-K_m\dot{\theta}_l - R_m i_{m,l} - u_{m,l} - K_m\dot{\theta}_r - R_m i_{m,r} - u_{m,r}\right) - \frac{r_m}{2}\frac{1}{L_m}\left(-K_m(\dot{\theta}_l - \dot{\theta}_r) - R_m(i_{m,l} - i_{m,r}) + u_{m,l} - u_{m,r}\right) = -\frac{K_m}{L_m}\dot{x} - \frac{R_m}{L_m}i_x + \frac{1}{L_m}u_x$

$\frac{di_y}{dt} = -\frac{K_m}{L_m}\dot{y} - \frac{R_m}{L_m}i_y + \frac{1}{L_m}u_y$

$$\Rightarrow \dot{\vec{x}}_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{2\mu_1+\mu_2}{J} & \frac{K_m}{J} \\ 0 & -\frac{K_m}{L_m} & -\frac{R_m}{L_m} \end{bmatrix}\begin{bmatrix} x \\ \dot{x} \\ i_x \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix} u_x \qquad \dot{\vec{x}}_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{2\mu_1+\mu_2}{J} & \frac{K_m}{J} \\ 0 & -\frac{K_m}{L_m} & -\frac{R_m}{L_m} \end{bmatrix}\begin{bmatrix} y \\ \dot{y} \\ i_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix} u_y$$

Q1.4 *Drawing shapes (1 pt.)*

Design input signals such that the tool head follows an elliptic path. Draw another closed shape of you choosing (i.e. a circle, a symbol from poker cards, ... use your imagination and be original!).

Show the working of your model through plots!

for ellipsoid $x = a\cos(\omega t)$ $\xrightarrow{\frac{d}{dt}}$ $\dot{x} = -a\omega\sin(\omega t)$ $\xrightarrow{\frac{d}{dt}}$ $\ddot{x} = -a\omega^2\cos(\omega t)$ $\xrightarrow{\frac{d}{dt}}$ $\dddot{x} = a\omega^3\sin(\omega t)$

$y = b\sin(\omega t)$ $\dot{y} = b\omega\cos(\omega t)$ $\ddot{y} = -b\omega^2\sin(\omega t)$ $\dddot{y} = -b\omega^3\cos(\omega t)$

from these differential equations line 2:

$\ddot{x} = -\frac{2\mu_1+\mu_2}{J}\dot{x} + \frac{K_m}{J}i_x \Rightarrow i_x = \frac{J}{K_m}\ddot{x} + \frac{2\mu_1+\mu_2}{K_m}\dot{x} \xrightarrow{\frac{d}{dt}} \frac{di_x}{dt} = \frac{J}{K_m}\dddot{x} + \frac{2\mu_1+\mu_2}{K_m}\ddot{x}$

$\ddot{y} = -\frac{2\mu_1+\mu_2}{J}\dot{y} + \frac{K_m}{J}i_y \Rightarrow i_y = \frac{J}{K_m}\ddot{y} + \frac{2\mu_1+\mu_2}{K_m}\dot{y} \xrightarrow{\frac{d}{dt}} \frac{di_y}{dt} = \frac{J}{K_m}\dddot{y} + \frac{2\mu_1+\mu_2}{K_m}\ddot{y}$

from 3rd lines

$\frac{di_x}{dt} = -\frac{K_m}{L_m}\dot{x} - \frac{R_m}{L_m}i_x + \frac{1}{L_m}u_x$

$\frac{di_y}{dt} = -\frac{K_m}{L_m}\dot{y} - \frac{R_m}{L_m}i_y + \frac{1}{L_m}u_y$

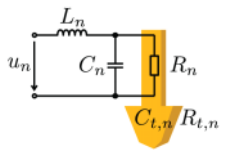
$\Rightarrow$

$u_x = L_m\frac{di_x}{dt} + K_m\dot{x} + R_m i_x + \frac{L_m J}{K_m}a\omega^3\sin(\omega t) - \frac{L_m(2\mu_1+\mu_2)}{K_m}a\omega^2\cos(\omega t) - K_m a\omega\sin(\omega t) - \frac{R_m J}{K_m}a\omega^2\cos(\omega t) - \frac{R_m(2\mu_1+\mu_2)}{K_m}a\omega\sin(\omega t) = \left[\frac{L_m J}{K_m}a\omega^3 - K_m a\omega - \frac{R_m(2\mu_1+\mu_2)}{K_m}a\omega\right]\sin(\omega t) + \left[-\frac{L_m(2\mu_1+\mu_2)}{K_m}a\omega^2 - \frac{R_m J}{K_m}a\omega^2\right]\cos(\omega t)$

$u_y = L_m\frac{di_y}{dt} + K_m\dot{y} + R_m i_y + \frac{L_m J}{K_m}b\omega^3\cos(\omega t) - \frac{L_m(2\mu_1+\mu_2)}{K_m}b\omega^2\sin(\omega t) + K_m b\omega\cos(\omega t) - \frac{R_m J}{K_m}b\omega^2\sin(\omega t) - \frac{R_m(2\mu_1+\mu_2)}{K_m}b\omega\cos(\omega t) = \left[-\frac{L_m J}{K_m}b\omega^3 + K_m b\omega + \frac{R_m(2\mu_1+\mu_2)}{K_m}b\omega\right]\cos(\omega t) + \left[-\frac{L_m(2\mu_1+\mu_2)}{K_m}b\omega^2 - \frac{R_m J}{K_m}b\omega^2\right]\sin(\omega t)$

Q1.5 *Model the Nozzle (1 pt.)*

The nozzle is heated by a resistor and is subject to heat exchange with the environment and the filament. Input to the circuit is a controllable voltage. Model the governing differential equation(s). At the



end, you should have a state-space discription of the form

$$\dot{\boldsymbol{x}}_n = f_n(\boldsymbol{x}_n, u_n) \tag{5}$$

with $\boldsymbol{x} = [T_n, \cdots]^\top$. You may combine the thermal loss to the environment and the filament into one loss with respect to the environmental temperature T_e, which you can choose.

$u_n = L_n\frac{di_n}{dt} + u_c \Rightarrow \frac{di_n}{dt} = \frac{u_n - u_c}{L_n}$

$C_n\frac{du_c}{dt} = i_c$ with $i_c = i_n - \frac{u_c}{R_n}$ $\Rightarrow \frac{du_c}{dt} = \frac{i_n}{C_n} - \frac{u_c}{R_n C_n}$

$W = C_{t,n}\dot{T}_n \qquad P_{in} = \frac{u_c^2}{R_n} \qquad P_{out} = \frac{T_n - T_e}{R_{t,n}}$

$W = P_{in} - P_{out} = \frac{u_c^2}{R_n} - \frac{T_n - T_e}{R_{t,n}} = C_{t,n}\dot{T}_n \Rightarrow \dot{T}_n = \frac{u_c^2}{R_n C_{t,n}} - \frac{T_n - T_e}{R_{t,n}C_{t,n}}$

$\vec{x}_n = [T_n, u_c, i_n]$

$$\dot{\vec{x}}_n = \begin{bmatrix} \frac{u_c^2}{R_n C_{t,n}} - \frac{T_n - T_e}{R_{t,n}C_{t,n}} \\ \frac{i_n}{C_n} - \frac{u_c}{R_n C_n} \\ \frac{u_n - u_c}{L_n} \end{bmatrix}$$